FEB 27 1981

Testimony: Transcript - Dec 12, 1978

SWD: b. 1913
age 65 left IBM 1975 on 62nd birthday ←

mgr of STRETCH - 1956-1961 (5yrs)
Yorktown — 1961-1966
Poughkeepsie — 1966-1975

P85527 -- burden - cost of level of transistors
need to have cost of level borne

529 -- transistor drives for cores.

Jan '56 - Project STRETCH Mr Pelusa

532 - desided to go to 400 cycle

534 - memory much slower than logical elements ←

536 - 7090 - lead to Goldstead
7080
Price Reduction ann, April 1961 ←

538 - costs of STRETCH

541 - interleaving memories

550 - differentiez - transistor speeds
- complexity

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553 - overlap of memories etc

555 - criticisms ←

559 - "hardwired" vs stored program

565 - re The 650

573 - "share of Market"

574 - date IBM was formed --

580 - tape processing machine

583 - 701

587 - scientific problems same as other problems

589 - "oral History" Sophie in 1968

597 - business vs scientific

606 - for "impeachment purposes"

613 - MANAC

615 - HG Kolsky "I hired him" ←

621 - "programming a machine"

626 - research or development laboratory

630 - allocating costs

693 – Livermore proposal "2-mega cycle"
694 – market estimate for STRETCH ? 37 machines ?
660 – proposal : "100 times greater"
662 – "42 months delivery" [Nov 56 + 42 mos → May '60]
664 – $4, 3 M
665 – " HG Kolsky -- refs to other proposals "
666 – Why did LASL accept IBM proposal ?
667 – Cuthbert Hurd - relation to LASL proposal
677 – HGK - quote on competitors
678 – memory speeds
689 – 7090 + 709T
694 – " it was a year late "
Dec 13, 78 698 – input output comparts
701 – speed of core memories 2.24 μs
703 – disk 1M wds → 2 → 4
705 – printer 1000 → 600 lpm
707 – " Transistor circuits were slow by a factor of 2 " (20 ns not 10)
additional complexity to compensate,
709 – LASL claim for compensation -- late delivery
+ 750,000 for rental credits (L. Robinson) 1964
715 – late delivery of NSA machine .
722 – forecasts
723 " commercial STRETCH ? "
730 " impact "
737 other customers : Wea Co, British AEC, Fords, Livmore, Dahlgren, CEIR
742 reliability -
746 costs – prices quoted to other customers
751 Livermore machine –
755 S. Campbell - meeting at Livermore
766 STRETCH price reduced June 1961 from $13.5 M to $8 M
776 a $22M loss over 7 year period 1956-62
783 list of losses - $1.2M on each of 5 systems $18.7 M
789 a "financial fiasco" ?
792 HGK "Comments on AEC Computer Situation" Aug 20, '63
794 "7034" –
795 Superstretch – Teller
804 The 22 items from STRETCH
815 any deleberate limiting of STRETCH perf?
823 McWhirter letter - denote & "take stock options away"
Redacted 827 IBM Fellow

85831 Mr Watson's apology for having reduced the price of STRETCH
839 "Legacies of STRETCH"
842 punched cards vs stored program computers
846 studies of Comm Edison, Time-Life
851 The "Watson tapes"
869 scientific vs business computation
876 701 series + 702 series "monsters"
879 Was STRETCH profitable?
880 Mr Watson's statement was excluded from evidence
885 does STRETCH run 100 times faster?
898 The reasons for Mr. Watsons apology?
907 "STRETCH was enormously profitable"
913 IBM's share of the business?
915 HGK quote - "Trade press" -
921 remote devices
926 pretrial brief?
939 March 1972 Tenth anniversary to NSA